

ACCRA TECHNICAL UNIVERSITY

Department of Computer Science

BCS 404: Introduction to Data Science with Python

Project Report

Exploratory Data Analysis, Statistical Analysis and Machine Learning Using the Titanic Dataset

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1. Introduction

The Titanic disaster is one of the most widely used datasets in data science and machine learning because it contains information that can be used to predict passenger survival based on demographic and socioeconomic characteristics. This project applies data science techniques to analyze the Titanic dataset obtained from Kaggle.

The project aims to explore the dataset, clean and preprocess the data, visualize important trends, perform statistical analysis, and develop a machine learning model capable of predicting whether a passenger survived the disaster. Python libraries including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn were used throughout the project.

2. Dataset Description

The dataset used for this project is the Titanic dataset obtained from the Kaggle Machine Learning Competition. It contains records of passengers aboard the RMS Titanic and includes demographic, travel, and survival information.

Each row in the dataset represents one passenger, while each column describes one characteristic of that passenger.

The important variables include:

Variable	Description
PassengerId	Unique passenger identifier
Survived	Survival status (0 = Did not survive, 1 = Survived)
Pclass	Passenger class
Name	Passenger's name
Sex	Passenger's gender
Age	Passenger's age
SibSp	Number of siblings/spouses aboard
Parch	Number of parents/children aboard

Fare

Ticket
fare

Embarked

Port
where
the
passeng
er
boarded

3. Methodology

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This project followed a structured data science workflow consisting of data acquisition, data cleaning, exploratory data analysis, statistical analysis, and machine learning.

3.1 Data Acquisition

The Titanic dataset was downloaded from the Kaggle Titanic Machine Learning Competition and imported into Jupyter Notebook using the Pandas library. The dataset dimensions, column names, first five records, and data types were examined to understand the structure of the data before analysis.

The dataset was checked for missing values and duplicate records. Missing values in the **Age** column were replaced with the median age, while missing values in the **Embarked** column were replaced with the most frequent category (mode). The **Cabin** column contained a large number of missing values and was therefore excluded from the machine learning model. Duplicate records were checked and removed where necessary.

3.3 Data Visualization

Several visualizations were created to better understand the characteristics of the passengers. These included:

- Histogram of passenger ages
- Passenger class distribution
- Boxplot of age by passenger class
- Scatter plot of Age versus Fare
- Correlation heatmap
- Pairplot of selected numerical variables

These visualizations helped identify trends, distributions, and relationships among the variables.

3.4 Statistical Analysis

Descriptive statistics were computed for the numerical variables to summarize the dataset. Frequency distributions were used to examine categorical variables, while correlation analysis was performed to determine the strength and direction of relationships between numerical variables.

3.5 Machine Learning

A Logistic Regression classifier was developed to predict passenger survival. The predictor variables included passenger class, sex, age, number of siblings/spouses, number of parents/children, fare, and embarkation port. The dataset was divided into training and testing sets. The model was evaluated using Accuracy, Confusion Matrix, and Classification Report.

4. Results

This section include the outputs from your notebook.

4.1 Histogram of Passenger Ages

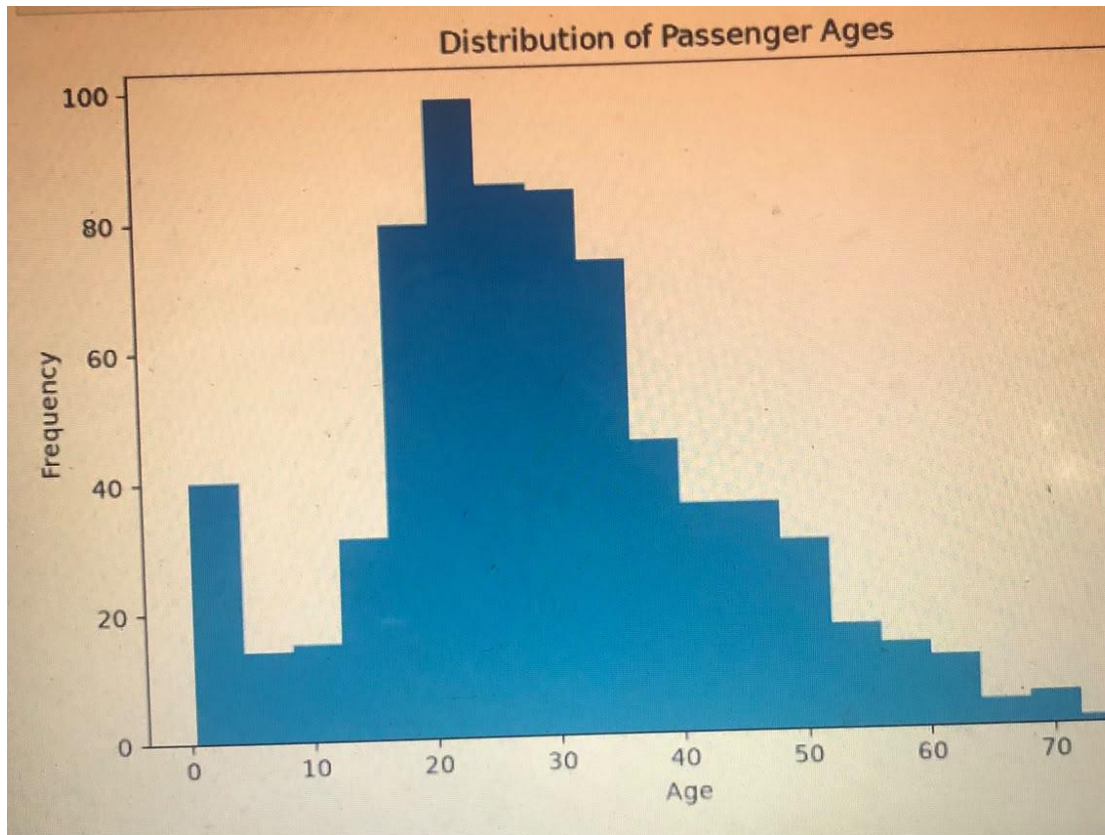


Figure 1: Histogram showing the distribution of passenger ages.

Interpretation

The histogram indicates that most passengers were between **20 and 40 years** of age. There were relatively fewer children and elderly passengers. The age distribution is slightly right-skewed, indicating that younger adults made up the majority of the passengers.

4.2 Passenger Class Distribution

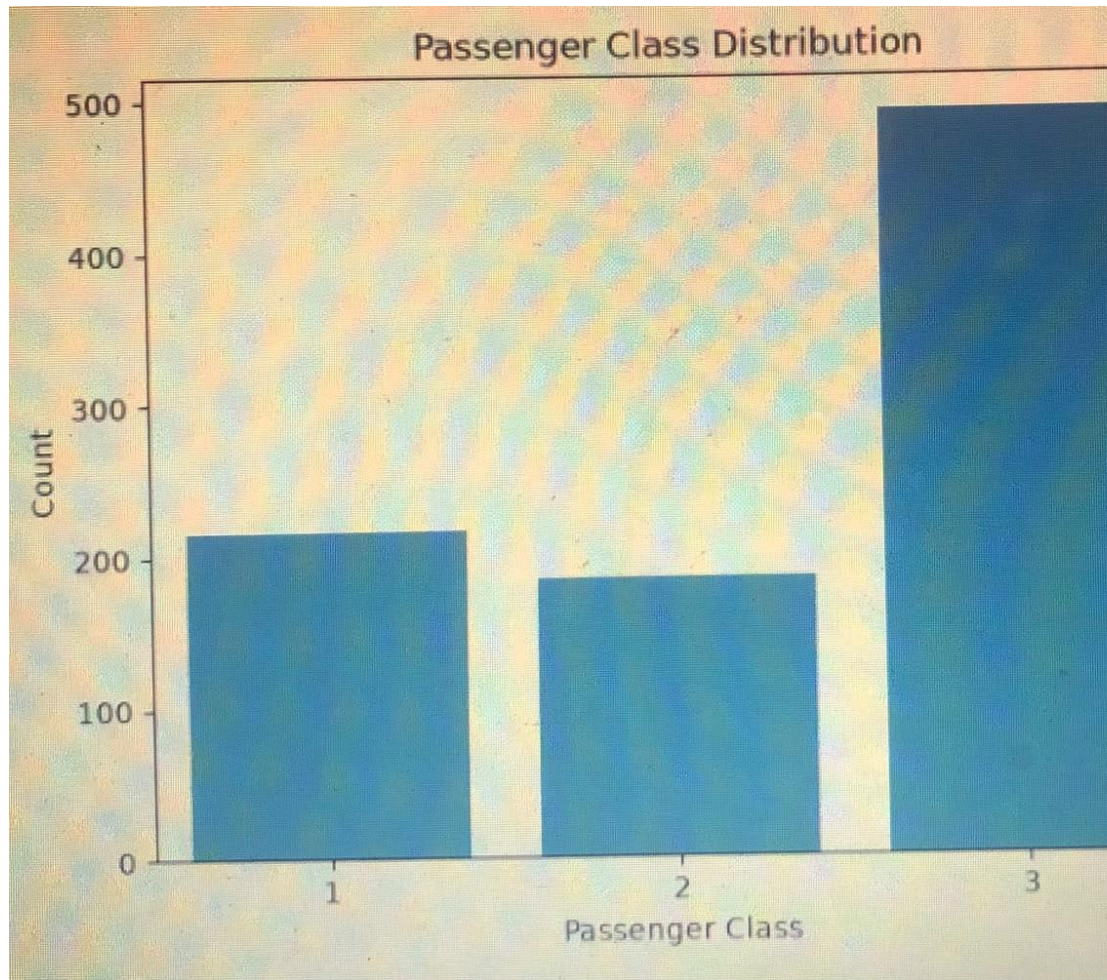


Figure 2: Bar chart showing the number of passengers in each travel class.

Interpretation

The bar chart shows that **Third Class (Pclass 3)** had the highest number of passengers, followed by First Class and Second Class. This suggests that most passengers traveled in the lower-priced class.

4.3 Boxplot of Age by Passenger Class

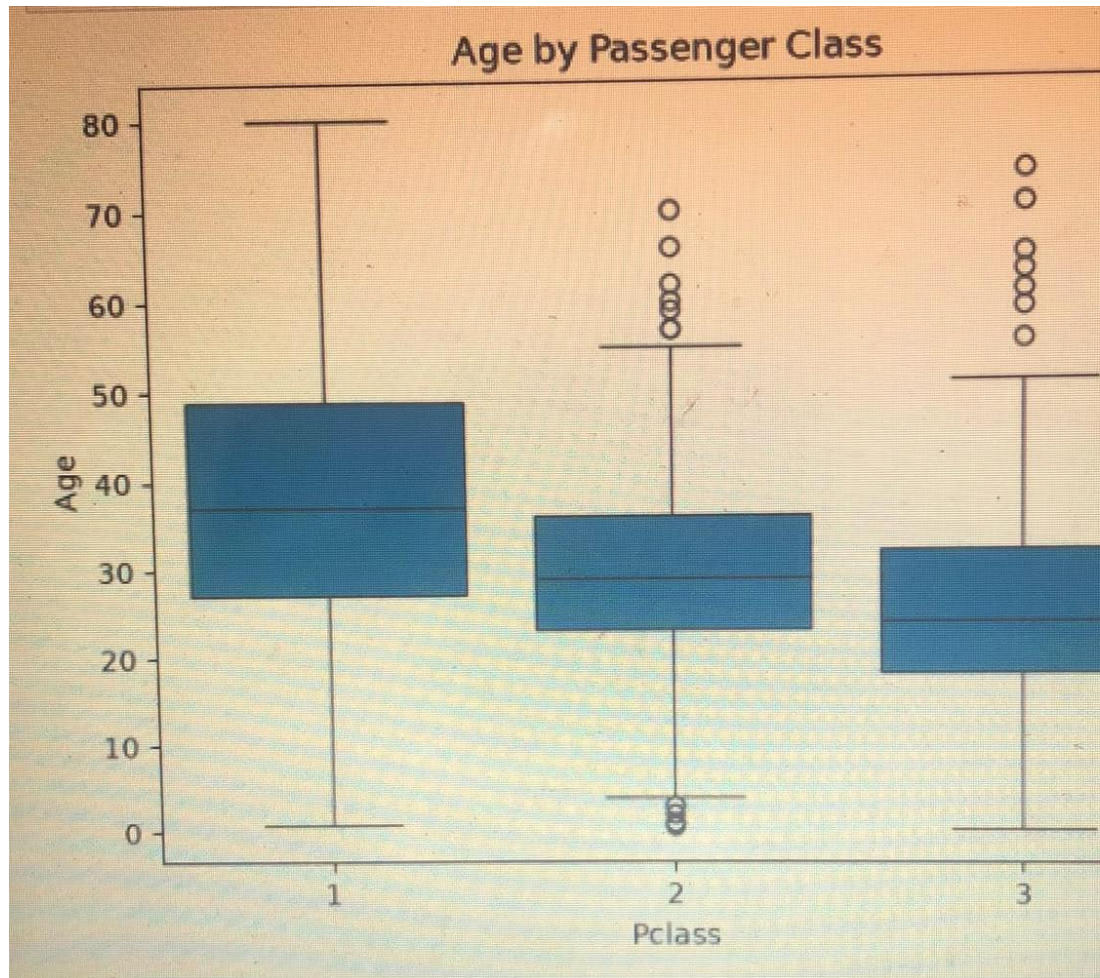


Figure 3: Boxplot comparing passenger age across travel classes.

Interpretation

The boxplot shows differences in the age distributions across the three passenger classes. First-class passengers generally had a higher median age, while third-class passengers included a wider range of ages and more younger passengers.

4.4 Scatter Plot of Age versus Fare

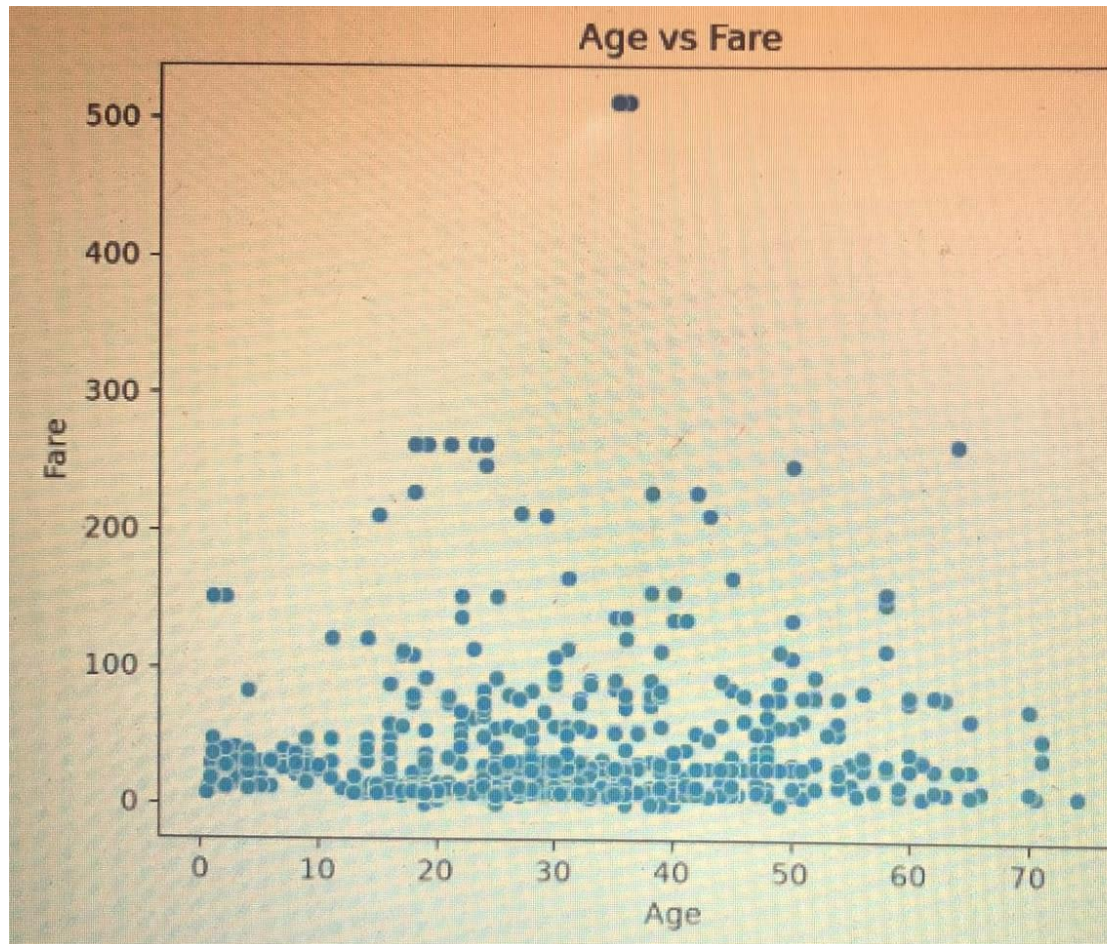


Figure 4: Scatter plot showing the relationship between passenger age and fare paid.

Interpretation

The scatter plot indicates that there is no strong linear relationship between passenger age and ticket fare. Passengers of different ages paid a wide range of fares, although some higher fares were associated with older passengers.

4.5 Correlation Heatmap

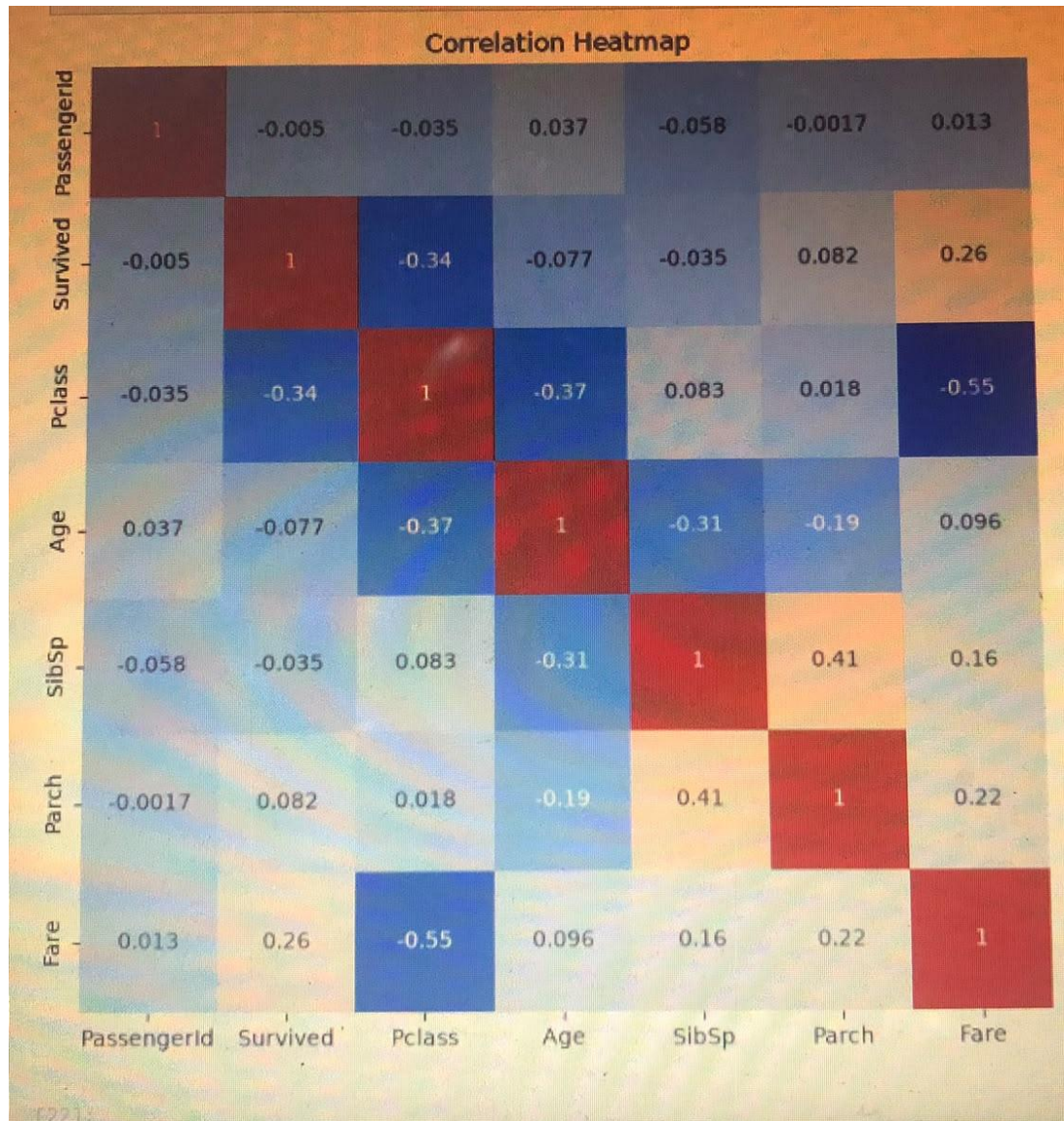


Figure 5: Correlation heatmap of numerical variables.

Interpretation

The heatmap illustrates the strength of relationships among numerical variables. Some variables show weak correlations, while others demonstrate moderate positive or negative relationships. This helps identify variables that may be useful for prediction.

4.6 Pairplot

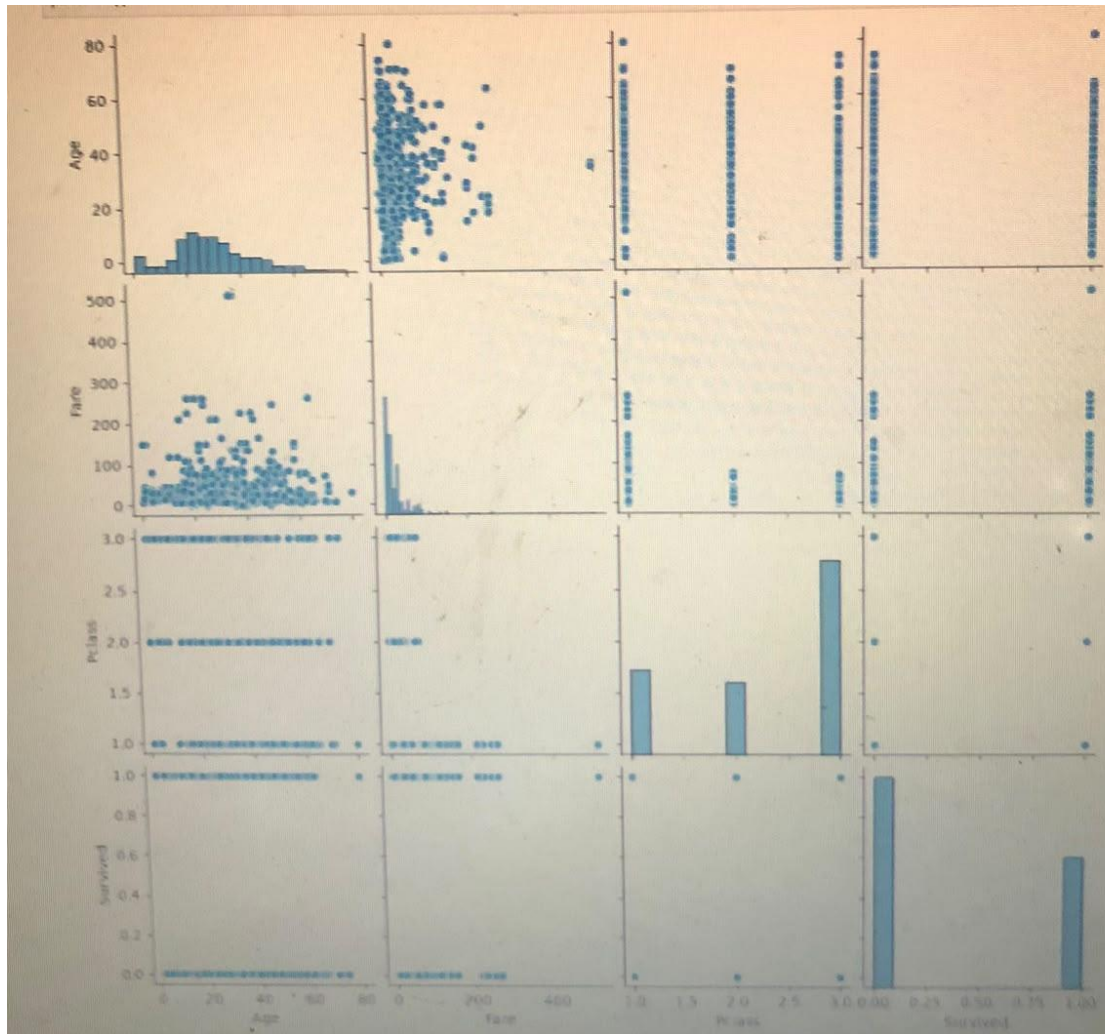


Figure 6: Pairplot of selected numerical variables.

Interpretation

The pairplot provides a visual comparison of relationships between numerical variables. It helps identify trends, clusters, and potential outliers that may influence the machine learning model.

5. Discussion

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The analysis of the Titanic dataset revealed several important patterns. The exploratory data analysis showed that passenger characteristics such as age, sex, passenger class, and fare varied across the dataset. Visualizations made it easier to understand these distributions and identify relationships between variables.

The data cleaning process improved the quality of the dataset by handling missing values and preparing the data for analysis. Categorical variables were converted into numerical values so that they could be used by the machine learning model.

A Logistic Regression model was developed to predict passenger survival. The model achieved an accuracy of approximately **73.18%**, indicating that it correctly classified about 73 out of every 100 passengers in the test dataset. This demonstrates that Logistic Regression can provide reasonable predictions using the selected features, although it is not perfect.

The confusion matrix and classification report provided additional insight into the model's performance by showing how well it identified passengers who survived and those who did not. Some prediction errors occurred because survival depended on several complex factors that were not fully captured by the available features.

Overall, the results demonstrate that data preprocessing, visualization, and statistical analysis play an important role in building effective machine learning models.

6. Conclusion

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This project successfully applied data science techniques to analyze the Titanic dataset. The dataset was cleaned, explored, and visualized to gain insights into passenger characteristics and survival patterns.

Statistical analysis helped summarize the data and identify relationships among variables. A Logistic Regression model was then developed to predict passenger survival. The model produced satisfactory performance, achieving an accuracy of approximately **73.18%** on the test dataset.

The project demonstrates how the data science process—from data acquisition and preprocessing to visualization and predictive modeling—can be used to solve real-world problems. Future work could improve prediction accuracy by testing more advanced machine learning algorithms such as Random Forest, Decision Trees, or Gradient Boosting.

7. References

1. Kaggle. (n.d.). *Titanic - Machine Learning from Disaster*. <https://www.kaggle.com/competitions/titanic>
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3. Pedregosa, F., et al. (2011). *Scikit-learn: Machine Learning in Python*. *Journal of Machine Learning Research*, 12, 2825–2830.
4. The Pandas Development Team. (2024). *Pandas Documentation*. <https://pandas.pydata.org/>

5. Hunter, J. D. (2007). *Matplotlib: A 2D Graphics Environment*. Computing in Science & Engineering, 9(3), 90–95.

8. Appendix

Appendix A: Python Source Code

Appendix A: The complete Python source code used for this project is provided in the submitted Jupyter Notebook file (**Titanic project.ipynb**).
